

Laser-induced fluorescence spectroscopy for combustion diagnostics

David R. Crosley
Gregory P. Smith
SRI International
Menlo Park, California 94025

Abstract. Laser-induced fluorescence can be used to detect with high sensitivity small molecular species, typically free radicals, which are the intermediates in the chemistry of combustion processes, the atmosphere, and plasmas. Using as examples recent work from our laboratory, we describe the laser spectroscopy and spectroscopically based collision studies needed for application of the laser-induced fluorescence techniques, with an emphasis on combustion diagnostics.

Keywords: fluorescence; laser-induced fluorescence; free radicals; collision studies; combustion diagnostics.

Optical Engineering 22(5), 545-553 (September/October 1983).

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1. INTRODUCTION

The technique of laser-induced fluorescence (LIF) of small, gas-phase molecules has found widespread application in the physical and engineering sciences. In addition to providing fundamental information on spectroscopic and collisional properties, it is proving useful as a diagnostic technique in several fields. LIF is especially suitable for the sensitive detection of a number of reactive intermediates in chemical networks, the most important example of which is OH. The areas in which LIF is increasingly used include the study of combustion phenomena, processes in the upper and lower atmosphere, the chemistry of plasmas, and laboratory measurements of chemical reaction rates.

Establishment of the ability to detect a given species with LIF requires a study of its spectroscopic characteristics. This often takes as a departure point previous, conventional (nonlaser) spectroscopic investigations, but involves further research to develop optimal detection strategies (choices of excitation and fluorescence wavelengths). This is especially true if there are potential interfering species. In many cases, further quantification of the LIF method is desirable; the extent and level of precision required vary with the species detected and processes studied. Here measurements are needed of lifetimes, transition probabilities, and a variety of collision phenomena. Finally, the development of special LIF variants such as two-photon detection requires additional spectroscopic detail.

We discuss in this paper the types of spectroscopic and collisional measurements needed to develop LIF as a diagnostic technique, using as illustrations recent experiments from our laboratory. (There are many other researchers and laboratories active in this field, and we emphasize that this paper is not to be construed as a review of work in the field in general.) Such studies form a significant portion of our research program, which has as its objective the development and application of laser methods (primarily LIF) for the understanding of the chemistry of a variety of processes. Much of our current orientation is toward combustion phenomena; the choice of species and characteristics for study and for discussion here is slanted toward that application. We note that of the nearly thirty combustion chemistry intermediates detectable by LIF, all were first studied by chemists or physicists in fundamental spectroscopic studies and not by the user community interested in applications. Hence, it is important to establish a coupling between the needs, present and anticipated, in the applied areas and an awareness of the current and possible capabilities of the fundamental studies.

2. LASER-INDUCED FLUORESCENCE AS A DIAGNOSTIC PROBE

2.1. LIF method

In laser-induced fluorescence¹ one tunes a laser so that its wavelength

Invited Paper F-105 received Mar. 3, 1983; revised manuscript received June 6, 1983; accepted for publication June 7, 1983; received by Managing Editor July 20, 1983.
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matches that of some absorption line of the species of interest. The molecules absorb the laser photons, becoming elevated to an electronically excited state from which they emit light (fluoresce). The fluorescent photons then form the detected signal. As the laser is scanned across a series of absorption lines, signal is produced each time a match occurs. Such a so-called excitation scan is depicted schematically in Fig. 1. It traces out what is essentially the absorption spectrum of the molecule, but with a significant difference compared to conventional absorption spectra. In LIF one detects not a small dip in some transmitted beam of light, but rather a positive signal on a null background. This yields much higher sensitivity; absorptions less than 10^{-6} per cm can produce readily measured signals.

The rate of production of molecules in the upper level pumped by the laser is given by

$$\frac{dN_e}{dt} = BIN_a, \quad (1)$$

where N_e and N_a are the number densities (cm^{-3}) in the pumped and absorbing levels, B is the Einstein absorption coefficient, and I is the laser intensity (W/cm^2). (We will not consider here line shape effects; BI in Eq. (1) represents actually the convolution of the laser and absorption lines over frequency, or $\int B(\nu)I(\nu)d\nu$.) N_a is related to the total ground-state density N_g by a Boltzmann distribution:

$$N_a = N_g G_a \exp(-E_a/kT), \quad (2)$$

where G_a and E_a are the degeneracy and energy of level a . The B -coefficient for a given v', v'' vibrational band and J', J'' rotational branch* is a product of three factors: a vibrational transition probability† $P_{v'v''}$, a rotational line strength $S_{J'J''}$, and an overall electronic transition probability, which is, in turn, inversely proportional to the radiative lifetime τ of the upper state. Thus,

$$B = \frac{P_{v'v''} S_{J'J''} \lambda^3}{8\pi h c \tau} \frac{G_e}{G_g}, \quad (3)$$

where λ is the transition wavelength, and the G 's are the electronic degeneracies of the excited and ground states.

The excited state then radiates. Under collision-free conditions, and when light from all the possible fluorescent transitions is collected and integrated over time, the fluorescence signal F is given by

$$F = cAN_e = cN_e/\tau, \quad (4)$$

where $A = \tau^{-1}$ is the Einstein emission coefficient. c contains all the factors of geometry (solid angle, probed volume), optical losses (filters or monochromator transmission, detector quantum efficiency), and electronic gain that can be separately calibrated. If a particular $J' \rightarrow J''$, $v' \rightarrow v''$ fluorescent transition is observed, the A of Eq. (4) becomes

$$A_{v'v''; J'J''} = P_{v'v''} S_{J'J''} / \tau. \quad (5)$$

Measurements of P , S , τ , and the energy levels of the molecule in question form the needed spectroscopic information for LIF diagnostic development.

At pressures of an atmosphere, a molecule that has been excited by the laser can suffer collisions during the time it resides in the upper state, before it radiates at the rate A characteristic of the electronic transition involved. For example, for the OH molecule in a flame at atmospheric pressure, only about 1 in each 1000 excited molecules emits a photon; the remainder are collisionally quenched back to the

*As in customary molecular spectroscopic notation, a single prime denotes the upper state and a double prime the ground state; also, when a transition is labeled, the upper state is always written first for both absorption and emission (e.g., the 6,0 band means $v' = 6$, $v'' = 0$).

†For the important OH molecule, but for no others of current practical significance, P varies with J' and J'' also, as discussed in Sec. 3.4.

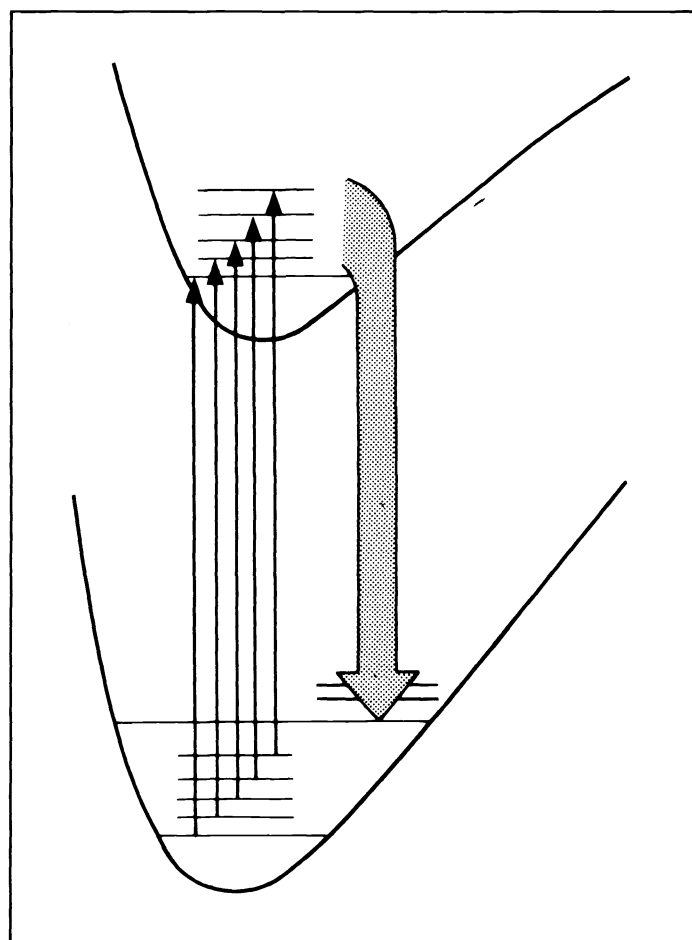


Fig. 1. Schematic representation of an excitation scan. The laser is tuned over various absorption lines (small arrows). Detection is achieved by measuring the total fluorescence intensity (large shaded arrow).

ground electronic state. In this case one has

$$F = cA\Phi N_e. \quad (6)$$

The fluorescence quantum yield Φ is related to A and the quenching rate Q by

$$\Phi = A/(A + Q). \quad (7)$$

Collisions can also be of the energy-transfer type, which move the molecule from the pumped level to other radiating levels within the upper state. In this case one must consider the state-to-state dependence of these collision processes together with the specific transitions monitored in fluorescence. That is, one may have an effective quantum yield differing from that in Eq. (7), as we shall discuss later. In any event, it is clear that the study of collisional effects² is also an important part of LIF development research.

2.2. LIF in combustion measurements

LIF, together with spontaneous and coherent Raman spectroscopy, forms a family of laser spectroscopic probes³ for the study of combustion, that is, methods providing concentrations and temperatures (through the population distribution over internal energy levels) of identifiable molecular species. The methods complement one another well in that the Raman techniques are the choices for measuring major species in a flame (the fuel, oxidant, main exhaust gases, and, in an air flame, N_2). LIF is the only method suitable for the detection of the chemical intermediates present at low concentration.

LIF has a number of attributes especially useful in combustion

studies. The laser can be focused and is generally pulsed, so that high spatial and temporal resolution are possible. Sample volumes of $<10^{-3} \text{ mm}^3$ can be defined, although 1 mm^3 is a more typically used value in many present studies. Most of the lasers used have pulse lengths of 10 ns, so that measurements average only over this interval. These attributes are important in atmospheric pressure flames, where significant concentration and temperature gradients can exist over spatial regions of the order of 1 mm, and under turbulent conditions, where conditions can change in times of the order of a few μs . The method is sensitive; for example, OH can be detected in 1 mm^3 volume at sub-part-per-billion concentrations, producing some 100 detected photons on a single 10 ns laser shot. Selectivity is obtained through the narrow bandwidth of the laser, which provides high spectral resolution. Even if two species absorb at the same bandwidth, they can often be distinguished by choosing different fluorescent wavelengths at which only one or the other emits. As with other optical probes, LIF is nonintrusive, so that it does not perturb the gas flows or chemical reactions. Additionally, it can be used in environments too hostile to permit the insertion of a physical probe device, such as a thermocouple or a sampling nozzle.

LIF is, in fact, the only method capable of providing sensitive, fast, spatially resolved measurements of chemical intermediates. It is, on the other hand, not general, as is mass spectrometry, for example. To be detectable with LIF, the molecule must possess a suitable electronic transition, at wavelengths accessible with available lasers, which also fluoresces. Fortunately, many important small combustion chemical intermediates fall into this category. Table I lists the "natural" combustion intermediates that have been detected with LIF in low pressure flows or cells and/or in flames. (Not included in the table are a number of metals, their oxides and halides, often obtained in flames only through seeding but sometimes found as contaminants, and many larger organic species, such as benzene, acetone, benzyl radicals, etc.)

For diagnostic applications, it is important to recognize that LIF measures the concentration of the ground electronic state of the species monitored, in contrast to the excited states seen directly from emission spectroscopy. Because the excited states are often produced through very different chemical reactions than ground states, the distributions can be very different, and only LIF furnishes information truly relevant to the overall chemistry.

Recently we have developed a new two-dimensional variant of LIF⁴ which promises numerous useful applications in flames and other systems. It is a fluorescence imaging technique that furnishes a planar map of the radical concentration throughout the flame on a single laser shot. Here, the laser is focused into a thin sheet of radiation (in our experiments on OH, it was 0.5 mm thick) and passed through the flame. Wherever the OH existed in the plane cut by the laser sheet within the flame, fluorescence resulted. This was focused at right angles onto the face of a two-dimensional vidicon tube, yielding an instantaneous picture of the OH concentration. Such a spatial correlation of the concentration is important in a turbulent system, where the conditions at the sampling point can vary drastically from shot to shot. The sensitivity for the OH fluorescence imaging was very promising; using 1.5 mJ of laser energy, ~2000 counts were obtained per mm^3 of sample volume at an OH

concentration of 700 ppm, on a single shot. The sensitivity for other species will vary with their spectroscopic characteristics.

2.3. Spectroscopic data needs

The particular needs for an LIF data base vary with the problem addressed by the method. In some cases, relatively crude knowledge may suffice. For example, a measurement of relative signal intensities for LIF in NH and OH in a $\text{CH}_4/\text{N}_2\text{O}$ flame was analyzed using the simple assumption that the quantum yield Φ for each was the same, given the total lack of quenching experiments on NH.² The surprising result that $[\text{NH}] \sim 0.04[\text{OH}]$ indicated a potentially important chemical role for nitrogen containing radicals in this flame; subsequent experiments⁵ have confirmed the high [NH] and found large amounts of CN and NCO as well.

Similarly, an imaging experiment with only a semiquantitative analysis may provide a great deal of information and stimulus to theorists approaching the problems of turbulent reactive flow; witness the theoretical advances following visualizations of turbulent structures using Schlieren methods. At the other extreme, one may wish to fit to a chemical model a detailed set of profiles of several radical species in a laminar flame. This may require much more information to assess precisely enough the relative signal strengths from different species over the changing thermal and collisional environments encountered through the flame.

The OH radical occupies a special position among molecules detectable by LIF. It is a ubiquitous and important intermediate in nearly any flame involving oxygen and hydrogen (as well as many other networks, such as atmospheric chemical processes). Its presence can be used to signify the occurrence of oxidation reactions. The needed laser wavelengths are experimentally very convenient, and its spectroscopic and collisional data base is far better established than that of any of the other molecules in Table I. It has served as a test species on which to try new methods (e.g., the imaging) and is generally the first species to search for when studying a new flame or probing a complex system. Despite the large spectroscopic effort devoted to OH, there remain some gaps in its data base, the closing of which is warranted by the premier importance of this radical.

3. SPECTROSCOPIC MEASUREMENTS UNDER COLLISION-FREE CONDITIONS

3.1. Production of radicals

The first step in performing laser spectroscopic experiments is to obtain an adequate concentration of the molecules to be studied. If it is a stable species, such as CO or SO_2 , a static cell may simply be filled at the desired pressure. Under some conditions, a static cell operated at specific conditions of temperature and pressure can lead to sufficient concentration of certain radical species; this method has been used for OH and S_2 . Most of the studies of free radicals, however, have been made in discharge flow systems. A parent species is dissociated in a microwave or radio frequency discharge at total pressures of the order of 1 Torr (generally the majority gas is an inert carrier such as He or Ar). In some cases the desired radical is directly produced in the discharge; N atoms may be produced from N_2 in this way, for example. In many cases the original radical reacts with another molecule to yield the species of interest. Thus, N atoms reacting with O_2 produce O atoms, and OH may be produced from $\text{H} + \text{NO}_2$. These reactions proceed rapidly to completion, so they can be used to produce known quantities of the radical through the addition of measured amounts of the stable reactant, O_2 or NO_2 . An attractive method for the production of many radicals is a discharge of CF_4 (in a He carrier) to produce F atoms; these then strip H atoms from a parent to produce radicals. We have used this method to produce the NCO radical from HNCO vapor, and it has been used for the production of alkoxy radicals (CH_3O , $\text{C}_2\text{H}_5\text{O}$, etc.).

Lasers can also be used for radical production in several ways. Single or multiphoton dissociation through an electronically excited state of the parent species (often with an excimer laser) has been used by several groups. A CO_2 laser can be used for multiphoton (infrared)

TABLE I. Combustion Intermediates Observed by LIF

O*, N, H, C, S
OH*
CH*, C ₂ *, CN*, CO*
NH*, NH ₂ *, NO*, NO ₂ *
NCO*, HCO, HNO
S ₂ *, SH*, SO*, SO ₂ *
CS, CS ₂
C ₂ O, C ₃ , CH ₃ O
C ₂ H ₂ , CH ₂ O, HCN

*Denotes that detection has been performed in a flame.

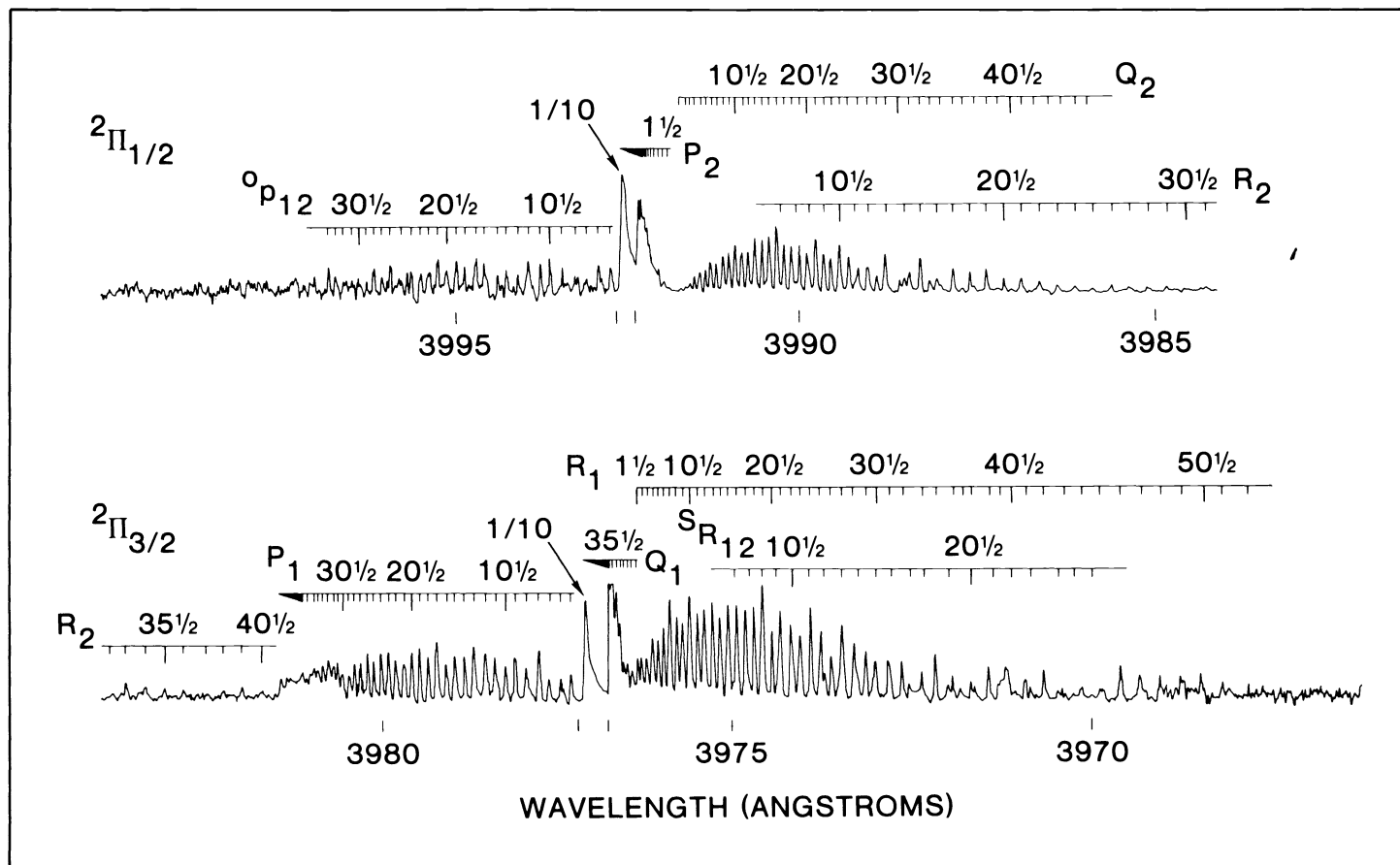


Fig. 2. A portion of an LIF excitation scan for the $A^2\Sigma^+(001)-X^2\Pi_i(000)$ band of NCO. Fluorescence was monitored at 431 ± 4 nm, in the 001-001 band.

dissociation of the parent to low-lying fragment states, or as a thermal gas heating source for dissociation. We have developed a laser pyrolysis/laser fluorescence method⁶ using a pulsed CO_2 laser to heat a gas sample. Here the infrared radiation is absorbed by SF_6 , and collisional energy transfer rapidly heats the system to a temperature chosen by the laser pulse energy and SF_6 pressure. Our experiments have used the radical precursor H_2O_2 ; this pyrolyzes to form OH radicals whose LIF is then detected with a dye laser timed to fire after the CO_2 laser. So far, only OH, NH_2 , and Fe atoms have been produced in this way although there are other possible candidate precursors and radicals.

The initial flow tube experiments are generally performed at low pressure and room temperature. This often yields larger signals than in a flame for two reasons: collisions do not reduce the quantum yield, and the population is distributed over a smaller number of internal levels than at high temperature. Using a flame as a radical source for initial search experiments also suffers from the disadvantages of more complex chemistry leading to interfering species, and the possible confinement of the species of interest to a very narrow spatial zone. Each of the molecules in Table I whose LIF has been observed in flames was first detected by LIF in a low pressure system.

3.2. Excitation scan studies

The next order of business after radical production is a search for the excitation scan of the molecule; that is, find the LIF signal. Again, all the molecules detectable with LIF have a common feature; such a search has used as a guide emission or absorption spectra previously obtained using conventional (nonlaser) methods. In many cases those measurements were made in flames, the medium in which the emission spectra of many radicals were first observed. Discharge emission spectra and the flash photolysis absorption studies of the 1950s and 1960s are a rich source of such data as well. Such starting points are most valuable, for an undirected search for sharp lines with a bandwidth of the same size as the laser bandwidth would be

very time-consuming.

Figure 2 exhibits an excitation scan for the NCO molecule.⁷ Here the excitation is from the 000 level* of the ground $X^2\Pi_i$ state to the 001 level of the excited $A^2\Sigma^+$ state. The energy levels involved are shown in Fig. 3. The ground-state level is split into two components separated by 98 cm^{-1} by a spin-orbit interaction,[†] and the excitations from each of the resulting $^2\Pi_{1/2}$ and $^2\Pi_{3/2}$ sublevels can be seen in the scan. Each component has associated with it an R ($\Delta J = J' - J'' = +1$), Q ($\Delta J = 0$), and P ($\Delta J = -1$) branch (as well as smaller O and S form branches with $\Delta N = \pm 2$), as marked on the figure. The assignment of these spectra followed from a detailed rotational analysis of absorption in NCO in a flash photolysis experiment.⁸ Excitation scans similar to Fig. 2 have also been performed pumping five other vibrational levels in the $A^2\Sigma^+$ state, as well as two vibrational levels of the $B^2\Pi_i$ state.

The relative intensity of each line within the band is given by the product of the line strength factor $S_{J,J'}$ and the fractional population N_a in the absorbing level. In this case the line strength factor may be calculated from knowledge of the angular momenta involved and the interactions among them (in this case, spin-orbit). Thus, once the lines have been assigned, the relative intensities may be used to obtain the rotational population distribution from which can be extracted the temperature from a plot of $\ln(N_a/G_a)$ versus E_a [Eq. (2)]. (If the system is not at thermal equilibrium so that a temperature is not defined, then the population distribution itself is of direct interest.)

Clearly, obtaining the total concentration from the intensity of a given line requires knowledge of the temperature so as to know the fractional population in the absorbing level [see Eq. (2)]. In cases where

*This notation signifies the three vibrations v_1, v_2, v_3 of this linear triatomic molecule. v_1 and v_3 are stretches, and v_2 is a twofold degenerate bend.

†Levels with $v_2 > 0$ have a more complex splitting pattern due to the vibrational angular momentum of the degenerate bend which interacts with the electronic orbital angular momentum, and the Renner-Teller splitting of the potential for a polyatomic Π state.

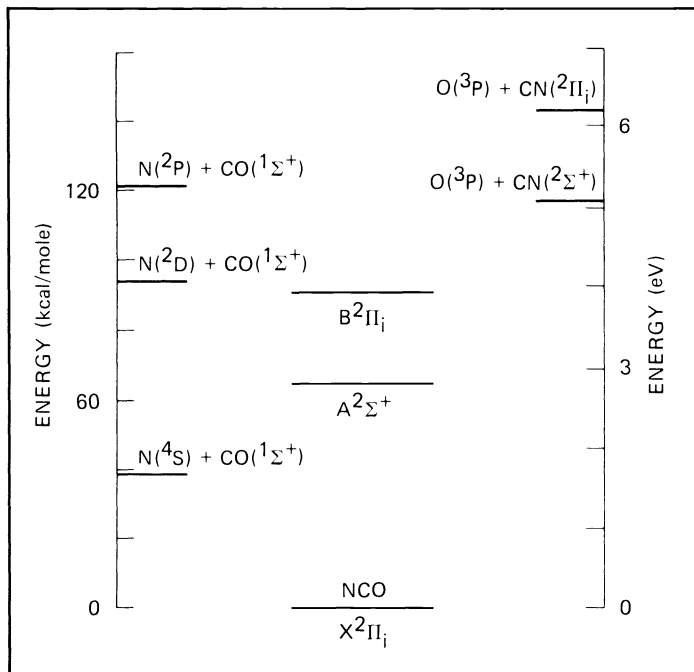


Fig. 3. Energy-level diagram for NCO showing the known electronic states and thermodynamic dissociation limits.

profiles are taken throughout a flame, it is often useful to choose a level for monitoring whose fractional population varies little over the temperatures of interest, that is, some J'' for which $\partial N_a / \partial T$ is small.

Much larger signal is obtained near the bandhead, where many lines pile up together so that the laser line overlaps the absorption from many levels at once (see Fig. 2). Excitation in a bandhead region may be quite useful, especially for low densities of the molecule studied, although analyzing the overlap to obtain the effective fractional population monitored may present problems for accurate measurements.

3.3. Lifetime measurements

The excited molecule radiates at a characteristic rate $k_r = 1/\tau$. This is directly related to the Einstein B- and A-coefficients [Eq. (3)], and its measurement serves as a useful way to determine absorption and emission strengths for the transition. For radical species it is often difficult to obtain a high enough concentration to perform precise absorption measurements; even when one can do so, accurate knowledge of the absolute concentration often presents problems. (To determine the strength of a given band, one must combine the lifetime measurements with determinations of relative $P_{v'v''}$, as discussed in the next section.)

For lifetimes longer than the pulse length of the laser, LIF furnishes a good method of measurement. Figure 4 shows the decay of fluorescence from the $v' = 1$ level of the $A^3\Pi_i$ state of the NH radical under collision-free conditions.⁹ Here a narrow electronic gate has been triggered initially just before the laser pulse, and its delay is scanned through the excitation (rising portion of the signal) and through the exponential decay. The data, fit to the exponential, furnish the lifetime. In the section on collisions, we will discuss quenching measurements made in this way at varying pressure of collision partner.

Lifetime measurements can furnish other information as well. In the case of NCO, the two different electronic states have very different radiative lifetimes: $\tau(A^2\Sigma^+) = 410$ ns; $\tau(B^2\Pi_i) = 63$ ns. The latter value was measured for the 000 level of the B-state. If one moves upward to the 100 level, only ~ 1000 cm^{-1} higher, the decay is indistinguishable from the laser pulse time dependence. Thus, the lifetime of this higher level is ≤ 10 ns. The marked decrease between these two vibrational levels is likely caused by the onset of a predissociation to the $N(2D) + CO(X^1\Sigma^+)$ limit (see Fig. 3). These lifetime

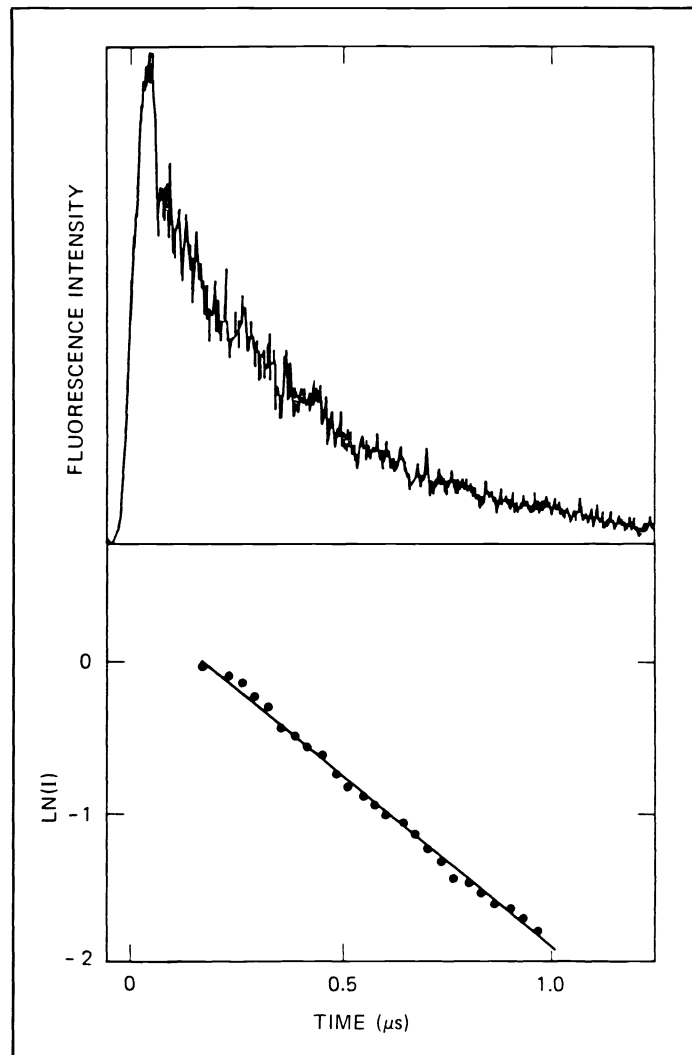


Fig. 4. Fluorescence decay of $NH A^3\Pi_i, v' = 1$, excited on the Q_{12} line in the (1,0) band at 305 nm and observed in the (1,1) band at 338 ± 20 nm, at collisionless pressures. The lower, logarithmic plot of the data gives a radiative lifetime of 440 ns.

measurements thus place this dissociation limit between these levels, 0.46 eV lower than determined in an earlier study of the photodissociation of HNCO. This new determination of the dissociation limit¹⁰ has as a consequence a higher heat of formation of both NCO and HNCO compared to the previously accepted values. The differences, nearly 10 kcal/mol, have implications concerning the role of the NCO radical in flame chemistry.

3.4. Fluorescence scans

A schematic indication of a fluorescence scan is given in Fig. 5. Here the laser is parked on one excitation. The level pumped radiates to some set of ground-state levels, for example, a series of vibrational levels. A monochromator is scanned to measure these transitions as a function of fluorescence wavelength. Under conditions where energy-transfer collisions occur before the molecule radiates, other levels are populated and can be discerned by such a scan; this phenomenon will be discussed in Sec. 4.

From each measured separation of a fluorescence wavelength from that of the pumping laser, one can obtain the separation in energy of a terminal v'' level for the fluorescence from the energy of the initial absorbing level. Figure 6 shows a fluorescence scan⁷ resulting from pumping the 000 level of the $B^2\Pi_i$ state of NCO. The laser line is off the chart to the left (at 315 nm). As the monochromator scans to longer wavelengths, one detects fluorescence returning to

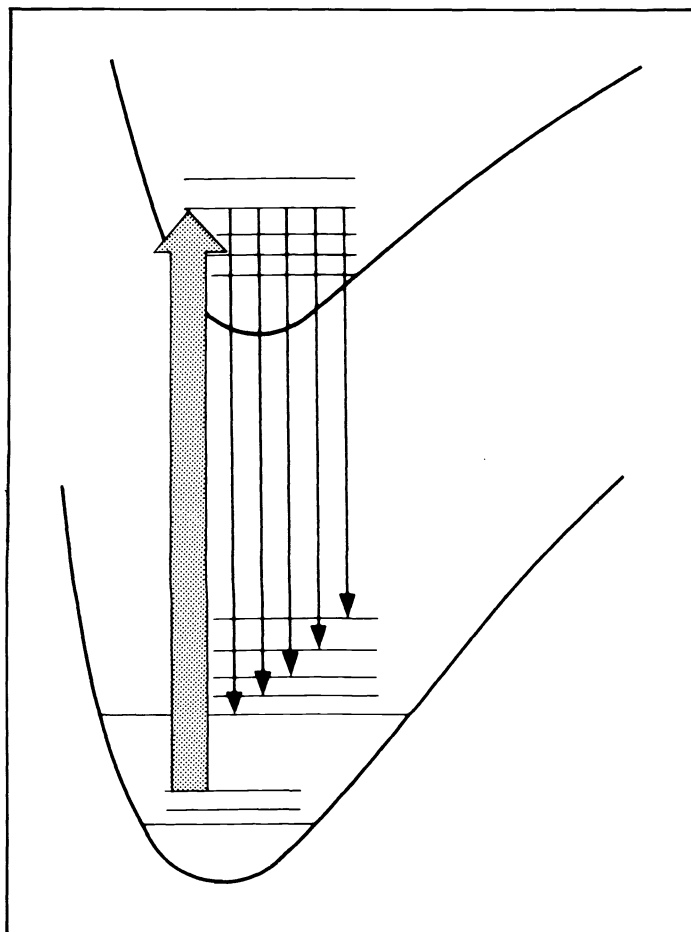


Fig. 5. Schematic representation of a fluorescence scan. The laser is tuned to a specific absorption line (shaded arrow) and the monochromator is scanned across the individual fluorescence emission lines from the excited level (small arrows).

higher v'' vibrational levels of the $X^2\Pi_1$ state. The more intense bands seen have one or more stretching vibrational quanta and $v''_2 = 0$; they are marked on the figure. Some weaker transitions (unmarked) involve $v''_2 = 2$ and 4. From a scan such as this (which extends further than is shown in the figure) one obtains the energies of the vibrational levels in ground-state NCO, which can be fit to a collection of harmonic and anharmonic vibrational constants. Previous information on NCO gas-phase spectroscopy came from absorption studies which probed only the 000 and 010 levels in the ground state, so that these results represent the first determination of these constants in the gas phase. The results may be used to understand the structure of the NCO molecule and to calculate its thermodynamic and thermochemical kinetic properties. From the standpoint of diagnostic measurements, knowledge of the wavelengths of the fluorescent transitions is needed in order to choose a suitable spectral region for detection.

Measurements of the intensities of the series of bands are also clearly important for the choice of an observation wavelength so as to maximize the available signal and/or avoid interferences. They are also needed for quantitative measurements of concentrations using LIF. We shall consider for simplicity a diatomic molecule with only one vibration although the concepts are applicable to larger species, such as NCO. For a particular $v'v''$ band, the relative intensity of fluorescence $P_{v'v''}$ is given by the integral

$$P_{v'v''} = \left| \int \psi_{v'}(r) R_e(r) \psi_{v''}(r) dr \right|^2 \quad (8)$$

over the two vibrational wave functions ψ and the electronic transition moment R_e , all expressed as functions of the internuclear distance r . $R_e(r)$ expresses the probability of the molecule making the

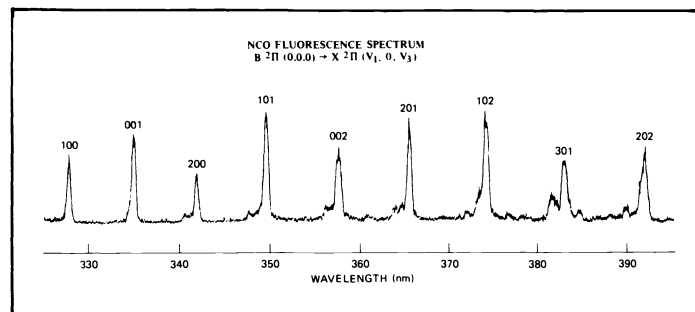


Fig. 6. Fluorescence spectrum from the 000 level of NCO $B^2\Pi_1$ excited at the R_1 head at 315 nm.

electronic transition as a function of internuclear distance. If it is constant, Eq. (8) reduces to

$$P_{v'v''} = R_e^2 \left| \int \psi_{v'} \psi_{v''} dr \right|^2 = R_e^2 q_{v'v''} \quad (9)$$

where $q_{v'v''}$ is simply the vibrational overlap integral, or Franck-Condon factor. For small species that have been well characterized spectroscopically, $q_{v'v''}$ is often calculable using potential curves for the pertinent electronic states constructed from the spectroscopic constants. If R_e varies with r , however, the relative $P_{v'v''}$ must be obtained from experimental intensity measurements. For larger molecules, one is generally better off using directly the empirical determinations of the $P_{v'v''}$.

An important example is the OH molecule, whose relative and absolute transition probabilities are reviewed in Ref. 11. For example, fluorescence scans¹² of the 0,0 and 0,1 bands show that $P_{01}/P_{00} = 0.004$ even though $q_{01}/q_{00} = 0.09$. The reason is that R_e decreases with increasing r , and the 0,1 band transition occurs at larger average values of r (1.22 Å) than does the 0,0 band (1.01 Å). A fit of the observed intensities for nine bands in each of OH¹² and OD,¹³ using calculated values of $q_{v'v''}$, showed that $R_e(r)$ could be fit to a linear form,

$$R_e(r) = c(1 - \rho r) \quad (10)$$

with $\rho = 0.75 \text{ Å}^{-1}$.

Transition probabilities in OH are unusual in another respect. The variation of $R_e(r)$ coupled with the high degree of centrifugal distortion present causes a significant variation in P with J . That is, the P values of Eqs. (3), (5), and (8) should be properly expressed as

$$P_{v'v'',J'J''} = \left| \int \psi_{v',J'}(r) R_e(r) \psi_{v'',J''}(r) dr \right|^2 \quad (11)$$

with the radial variation of the rotational wavefunction included. The form of $R_e(r)$ determined from the LIF fluorescence scans, given in Eq. (10), has been combined with accurate wave functions $\psi_{v,J}(r)$ from RKR potentials to calculate $P_{v'v'',J'J''}$ for all bands of the OH $A^2\Sigma^+ - X^2\Pi_1$ system which are of diagnostic interest.¹⁴

No other molecule is as well characterized as OH, and even where numerous studies exist for others, there can remain uncertainty in the results. An example is the C_2 molecule. A table of previously measured lifetime values of the $A^3\Pi_1$ state (upper level of the Swan bands) was given in the paper¹⁵ describing the first application of optically saturated LIF in combustion, because this value was needed to analyze the experimental data. A variation of more than a factor of seven, over 17 different measurements, can be seen.

4. COLLISIONAL EFFECTS

As noted earlier, the collisions which the molecule suffers while excited can have a profound influence on both the magnitude and spectral form of LIF signals. The collisions may be of a quenching nature, reducing the overall quantum yield Φ [Eqs. (6) and (7)]. Collisions can also cause energy transfer to other v', J' levels of the excited state. These collisionally transferred molecules will then fluoresce at different wave-

lengths and with different intensities compared to the originally pumped level. Thus, the effective quantum yield into the detector bandpass can be affected, particularly if it is narrow.

The subject of collisional effects on LIF measurements in flames has been reviewed with coverage of the literature through early 1981, and we refer the reader to that article² for a comprehensive discussion. Emphasized here are the fundamental methods used to study collisional effects, and studies performed since the appearance of Ref. 2.

4.1. Quenching

Quenching rate constants k_Q for individual collision partners are generally obtained from lifetime measurements. The time dependence of the excited-state population is given by

$$N_e(t) = N_e(t=0) \exp[-(k_r + k_Q n)t], \quad (12)$$

where $N_e(t=0)$ is the population initially created by the laser pulse, and n is the density of the added collision partner. A plot of the total decay rate versus pressure of added O_2 and N_2 for the 100 level of the $A^2\Sigma^+$ state of NCO is shown in Fig. 7; from the slopes we obtained k_Q for these gases.⁷ In such experiments it is essential to measure all of the excited-state fluorescence. Suppose, for example, a $v'=1$ level is pumped by the laser and the choice of fluorescence wavelengths is such that only emission from that level is detected. Then any collisional transfer to $v'=0$ will appear as quenching, and the slope of the plot will not give the true value of k_Q .

Lifetime measurements may be used only under conditions in which the decay time is longer than the laser pulse length, typically 10 ns for the lasers used in these experiments. At a high enough pressure, the total decay time becomes shorter than this, and the population $N(t)$ will reflect only the time dependence of the laser pulse. For some molecules k_r is already too fast, precluding quenching determinations from lifetimes. Under conditions where the number density of the molecules pumped by the laser can be controlled, fluorescence intensity measurements can yield quenching rates. An example is CO ,¹⁶ pumped by two photons at 289 nm to the $v'=3$ level of the $A^1\Pi$ state, which has a lifetime of 10 ns. The total fluorescence intensity F , integrated over laser pulse and decay time, is compared for a given CO density, with (F) and without (F_0) an added gas at density n :

$$\frac{F_0}{F} = \frac{k_r + k_Q^{CO} n_{CO} + k_Q n}{k_r + k_Q^{CO} n_{CO}}. \quad (13)$$

A plot of this ratio for N_2 as the collision partner is given in Fig. 8; runs at three separate n_{CO} values, normalized for comparison, are shown. Both k_r and k_Q^{CO} must be separately measured to determine k_Q for N_2 .

Of the molecules in Table I, k_Q is known for a variety of collision partners only for OH. Even here, all the previous direct measurements have been made at room temperature, where the flow systems used for the radical production operate most easily. Extensions of these rate constants to flame conditions have been made by assuming that the cross section σ_Q is constant, so that the only temperature variation in k_Q is $T^{1/2}$ from the velocity dependence. However, experiments on state-to-state energy transfer in $A^2\Sigma^+$ OH¹⁷ suggest that attractive forces play a role in its collisional phenomena. If this is the case, one expects that σ_Q will decrease with increasing temperature,¹⁸ so that k_Q will increase less rapidly than $T^{1/2}$.

Using our laser pyrolysis/laser fluorescence apparatus,⁶ we have made measurements of k_Q for OH at temperatures in the 1100 to 1400 K region for a variety of collision partners.¹⁹ The pulsed CO_2 laser radiation is absorbed by SF_6 , heating the gas mixture and pyrolyzing H_2O_2 to OH radicals; these are pumped by the dye laser to $v'=0$ of the A-state at a fixed time delay after the CO_2 pulse. Measurements of the fluorescence lifetime for ten different collision partners of importance in flames have been made and show in general that the cross section is indeed smaller than at room temperature. In particular, k_Q for N_2 is anomalously small, with a value of $1.2 \times 10^{-11} \text{ cm}^3 \cdot \text{s}^{-1}$. This is much lower than previously assumed at flame

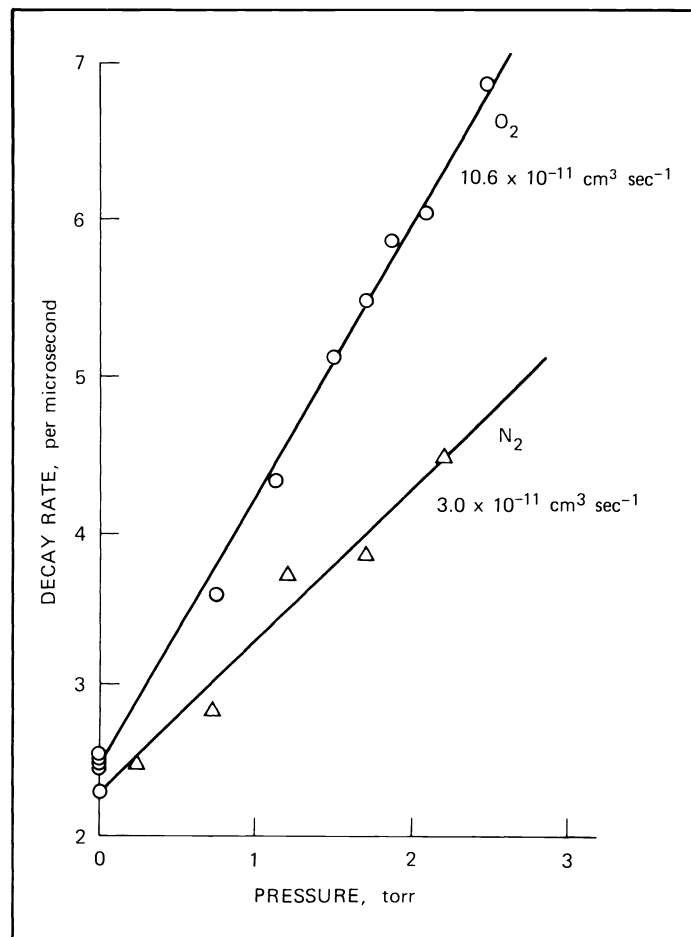


Fig. 7. Plots of the fluorescence decay rate for NCO $A^2\Sigma^+$ observed on the $\Delta v = 0$ bands after excitation of the 001 level as a function of added pressure of oxygen and nitrogen. Lines give the least-squares quenching rate constants.

temperatures and has important implications for LIF signal strengths in air-based flames.

The results for the other collisional partners have been examined¹⁹ in the context of theoretical approaches which express σ_Q in terms of attractive well depths between the excited OH and its collision partner,¹⁸ and a combination of multipole interactions.²⁰ The trends in σ_Q with the characteristics of the collision partner confirm the conclusion from the temperature comparison that attractive forces are important in the quenching process. Further examination of OH and other excited radicals will be important not only for LIF diagnostic development but also to establish a fundamental understanding of collisional mechanisms.

Results for k_Q for a variety of gases, both directly measured and estimated from these correlations, have been combined with profiles of major species (the predominant quenchers) published for a number of low pressure flames. The results²¹ indicate little variation in the total quenching rate Q through the flame. This is in agreement with direct determinations of Q by lifetime measurements in low pressure flames for both OH²² and CH,²³ which show a relatively constant value (within 20%) through the flame fronts.

4.2. Energy transfer

Collisions may also transfer the excited molecule among $v'J'$ levels of the upper state. This phenomenon is studied by making fluorescence scans at controlled pressures of added gas. An example²⁴ for S_2 is shown in Fig. 9. Here changes in the fluorescence spectrum* from the

*The actual scan from which this figure was made employed as an excitation source a Zn atomic lamp, one of whose lines coincidentally overlapped an S_2 line, but the experiment could be performed (actually more easily) using laser excitation.

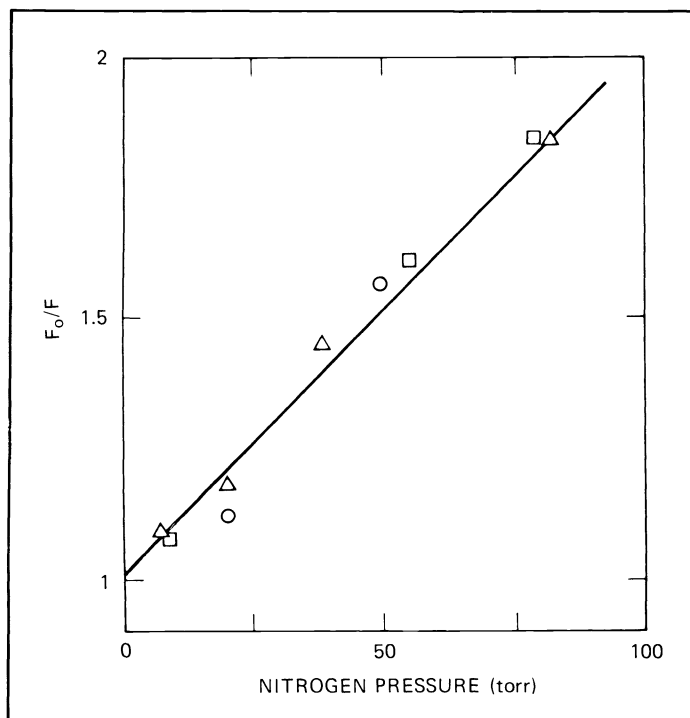


Fig. 8. Comparative plot of total fluorescence intensity vs N_2 pressure for $CO A^1\Pi$ excited to $v' = 3$. Data are shown for three separate runs. The ordinate ratio F_0/F , given by Eq. (13) in the text, is the fluorescence intensity without and with added N_2 . The CO pressures are: circles and squares, 20.1 Torr; triangles, 23.8 Torr.

$B^3\Sigma_u^-$ state, upon addition of successively higher pressures of Ar, are seen. First rotational transfer occurs within the initially pumped $v' = 4$ level, and then the molecule undergoes vibrational relaxation with the population piling up largely in $v' = 0$ and 1.

The degree to which such thermalization within the excited state occurs depends on the relative rates of quenching Q and rotational (R) or vibrational (V) energy transfer. If $Q \ll R, V$, the molecule will relax before being quenched. However, if $Q \gg R, V$, then the single level pumped by the laser will not be transferred before being quenched from the excited state. The resulting upper-state population distribution will be highly peaked and nonthermal, a situation that has been termed arrested relaxation, or frozen excitation. Studies made in flames indicate an intermediate situation, so that partial relaxation occurs. Fluorescence scans have been made after pumping several levels in OH, which show rotational^{11,25} and vibrational relaxation,²⁶ and both effects were seen for a single level pump in CN.⁵ The results on OH exhibit distinct state-dependent features^{11,26}; that is, both the rate and magnitude of the transfer are dependent on the initial level pumped. These effects must be accounted for to obtain quantitative determinations of temperatures and concentrations from LIF measurements.²⁷ Thus, knowledge of state-to-state transfer propensities is needed for LIF diagnostic development, especially for the important OH molecule.

Such experiments are performed much in the manner illustrated in Fig. 9, using the spectral resolution necessary to distinguish rotational and/or vibrational levels, depending on which phenomenon is to be studied. Except at the lowest pressures, multiple collisions can occur (as evident in Fig. 9), and these effects must be incorporated into the data analysis to obtain true state-specific rates. In general, it is found that multiple-quantum ($|\Delta v|$ or $|\Delta J| > 1$) jumps can occur on single collisions although the probability falls off with increasing separation of the initial and final levels.

4.3. Polarization phenomena

Most laser radiation is linearly polarized; that is, its electric vector

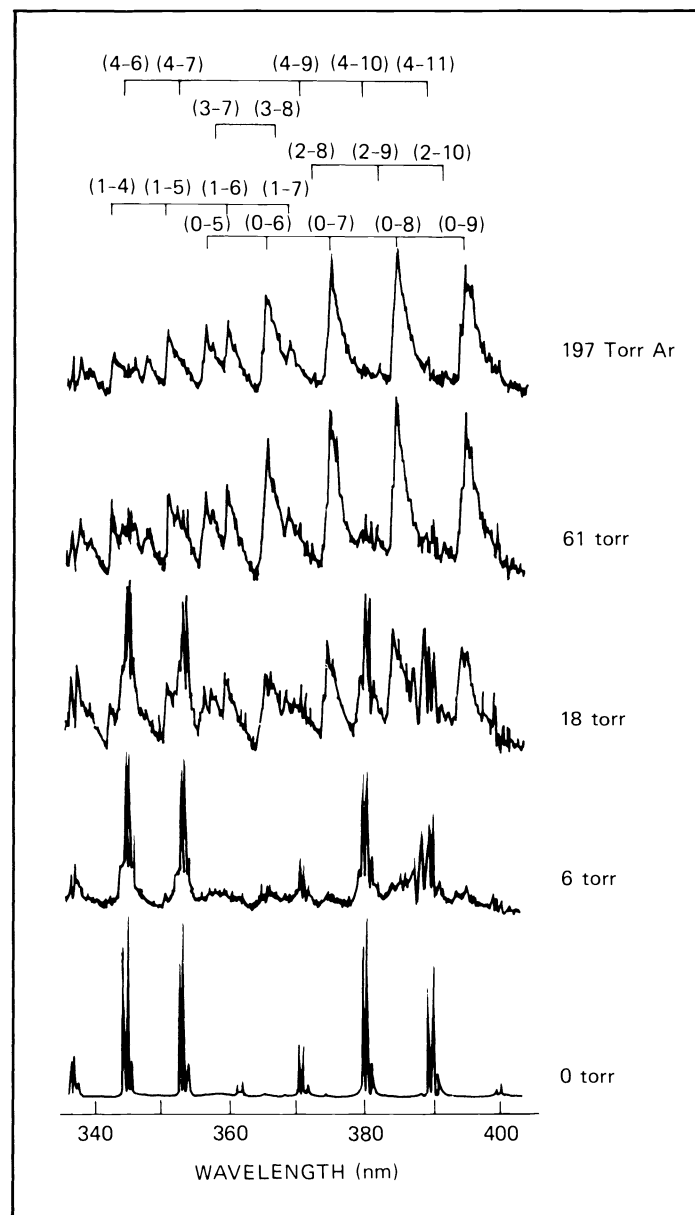


Fig. 9. Fluorescence spectra of $S_2 B^3\Sigma_u^-$ initially excited in $v' = 4$, $J' = 41$, at various pressures of added argon gas. As pressure is increased, rotational energy transfer first populates other J' levels in $v' = 4$, broadening the bands. At higher pressures, vibrational energy transfer populates other v' levels. Other bands with lower v' appear as those with $v' = 4$ decrease in intensity.

has a particular relationship to laboratory-fixed axes. This electric field interacts with the transition dipole moment in the molecule, which in turn bears a fixed relationship to the internal molecular axes. The result is that the excited-state molecule has a definite orientation in space (described quantum mechanically by a particular distribution over over m_j magnetic sublevels), and the fluorescent emission from the isolated molecule will have an anisotropic spatial distribution. This is manifested as a polarization of the fluorescence which varies with viewing direction. For a simple dipole oscillator this anisotropy is total; that is, no light is emitted in the direction of the original laser electric field. In the case of a molecule, the rotation reduces the degree of polarization. However, because the rotational angular momentum, which is a constant of the motion, bears a definite relationship to the molecular axes, a significant degree of polarization can remain.

If, as in a flame, collisions take place before the molecule emits, the degree of polarization will be reduced still further. Collisions

causing vibrational and rotational transfer can at the same time scramble the orientation; in addition, elastic depolarizing collisions (at a rate D) can alter the orientation of the molecule in space while leaving it in the initially pumped level. The degree of resulting polarization of light emitted by the pumped level is then governed by the ratio $D/(Q + V + R)$, while that of light emitted by collisionally populated levels is dependent on the amount of disorientation that occurs during such a process.

The first direct evidence of polarization of LIF in flames was obtained by the observation of apparent differences in rotational line strengths in OH,²⁸ in qualitative accord with earlier calculated predictions.²⁹ We have made measurements of the degree of polarization of OH LIF in the burnt gases of a series of methane/oxygen flames in N_2 , Ar, or He diluent.³⁰ Surprisingly, the degree of polarization is very high, nearly that which would be found for a collision-free case; this indicates that $D \ll Q + R + V$. Even more surprising is the fact that the polarization of the fluorescence emitted by nearby levels populated by rotational energy transfer is also quite high; that is, such collisions alter the magnitude of the rotational angular momentum without randomizing its direction in space. Vibrational transfer collisions, on the other hand, scramble the orientation, in accord with a picture of attractive forces and long-lived collisions as the responsible factors. Polarization has also now been observed in OH with LIF under optical saturation conditions.³¹

Calculations^{29,30} indicate that these effects are most pronounced when the fluorescence is rotationally resolved. Since P- and R-branches exhibit polarization opposite to that of the Q-branches, a sum over the entire band reduces these effects to a few percent.

5. CONCLUSIONS

In this paper we have attempted to provide a feel for the necessary experiments that make up the spectroscopic and collisional data base needed for quantitative detection of reaction intermediates in flames using LIF. Such research is being performed in our laboratory and elsewhere, and we reemphasize that this is not intended as a comprehensive review. This data base continues to grow, especially for OH and to a lesser degree for the other species listed in Table I. As noted earlier, it is important that the needs of the user community and the capabilities of the basic spectroscopic researchers be mutually communicated, so as to direct future efforts in this area as fruitfully as possible.

6. ACKNOWLEDGMENTS

The experiments described herein which were carried out in our laboratories were the result of collaboration with a number of colleagues. It is our pleasure to thank Tom Caughey, Irv Chidsey, Paul Doherty, Mark Dyer, Paul Fairchild, Russ Lengel, and Brian Sulli-

van for their invaluable contributions. We are also grateful for the support for this research, which is provided by the U.S. Army Research Office, the National Science Foundation, the Department of Energy, the Air Force Wright Aeronautical Laboratories, and the National Aeronautics and Space Administration.

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